Exact Character Reductions and a Restricted Fourth-Moment Concentration Route for Arithmetic Quantum Cat Maps

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Scientific status. This manuscript is a research draft, not a proof of the general Bohigas–Giannoni–Schmit conjecture. Exact finite-field reductions and several finite combinatorial identities below are proved and independently verified. A restricted fourth-moment concentration theorem in the regime $K = o(N^{1/4})$ has a complete proof *route* assembled from exact algebraic reductions and standard ℓ-adic inputs, but is not promoted to a final theorem until the algebraic computer-assisted lemmas, projective models, and the cited sheaf-theoretic hypotheses have passed a line-by-line publication audit.

Abstract

We study arithmetic moment structures arising from quantized hyperbolic cat maps at split prime levels. The main technical contribution is a replacement of an earlier, invalid Möbius-fiber argument by an exact weighted pair-key geometry. For a multiplicative subgroup $H \leq \mathbf{F}_N^\times$ of order K and its annihilator family of characters, the complete fourth-moment twist sum collapses to one finite-field difference distribution $B(t)$. This gives exact formulas for the total $k = 4$ constraint variety, the fourth-moment Fourier transform, and an exact mean/variance duality. The lower-dimensional mixed term is transformed by a Cayley map into an additive-energy problem. For odd prime K, outside an explicit finite cyclotomic exceptional set of primes N, its energy is exactly $(K-1)(2K-5)$. The remaining off/off term is organized as a non-isotrivial elliptic pencil. Its j-map has degree 12, its generic parameter symmetries are classified, and the only N-scale reducible-fiber spikes occur at the already-controlled Cayley differences. These facts reduce a first rigorous concentration theorem to standard quasi-orthogonality for bounded-complexity elliptic trace sheaves, yielding the natural preliminary range $K = o(N^{1/4})$. We also record the computational certificates, failed historical routes, and the precise gap between fourth-moment concentration and a local BGS spacing theorem.

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1 Scope and relation to BGS

The BGS conjecture predicts random-matrix local spectral statistics for quantizations of classically chaotic systems after all relevant symmetries are removed. Arithmetic cat maps are unusually delicate because Hecke symmetries and finite-field algebra can dominate spectral and eigenfunction statistics. Consequently, neither quantum ergodicity nor a successful matrix-element moment calculation automatically implies GOE/GUE local spacing statistics.

The current project therefore separates three logical layers:

1. exact finite-field character and moment identities;
2. asymptotic concentration of the resulting finite-field moment objects;
3. a later local spectral-law step connecting those estimates to unfolded eigenphase statistics.

This paper concerns primarily the first two layers. The third remains open.

2 Finite-field setup and character conventions

Let N be an odd prime and let

$$H = \langle a \rangle \leq \mathbf{F}_N^\times, \qquad |H| = K, \qquad m = \frac{N-1}{K}.$$

Let $S = H^\perp$ be the annihilator subgroup of multiplicative characters, so $|S| = m$. If g is a primitive root modulo N and $\chi_j(g) = \exp(2\pi ij/(N-1))$, then

$$S = \{\chi_{Kj} : j \bmod m\}.$$

In particular the natural generator χ_K has order m, not K. This distinction matters in every perfect-power or divisor argument involving the twist family.

For $\chi, \psi \in S$ and additive frequency $q \in \mathbf{F}_N$, define the twisted Jacobi-type sum

$$J(\chi, \psi; q) = \sum_{x \in \mathbf{F}_N \setminus \{0,1\}} \overline{\chi(x)}\, \psi(x-1) e_N(qx), \qquad e_N(z) = \exp\left(\frac{2\pi iz}{N}\right).$$

The exact moment reductions below arise by expanding powers of J and using orthogonality over S.

3 The exact pair-key geometry

3.1 Pair coordinates

For ordered pairs $x, y \in \mathbf{F}_N \setminus \{0,1\}$ define

$$u = \frac{x}{y}, \qquad v = \frac{x-1}{y-1}, \qquad d = x - y.$$

The diagonal $x = y$ collapses to one key $(u, v) = (1, 1)$ with multiplicity $N - 2$ and $d = 0$. Off the diagonal, the map is a bijection onto

$$D_{\text{off}} = \{(u, v) \in (\mathbf{F}_N^\times \setminus \{1\})^2 : u \neq v\},$$

with inverse

$$y = \frac{1-v}{u-v}, \qquad x = \frac{u(1-v)}{u-v},$$

and exact difference

$$d(u, v) = \frac{(u-1)(v-1)}{v-u}. \tag{1}$$

Thus the collapsed table has $(N-2)(N-3)+1$ records: one weighted diagonal record and $(N-2)(N-3)$ unit-weight off-diagonal records.

3.2 The second-moment constraint

The second-moment constraint requires the off-diagonal key itself to lie in $H \times H$. Hence the number of constrained off-diagonal pairs is exactly

$$(K-1)(K-2).$$

This count is independent of all later $k = 4$ asymptotics and is checked directly in the active verification suite.

4 The fourth-moment difference distribution

Let a collapsed key record be $r = (u_r, v_r, d_r, n_r)$, where n_r is its multiplicity. Define the ordered coupled difference distribution

$$B(t) = \sum_{r,s} n_r n_s \, \mathbf{1}_H(u_r u_s) \, \mathbf{1}_H(v_r v_s) \, \mathbf{1}_{d_r + d_s = t}. \tag{2}$$

Write

$$V_4 = \sum_t B(t), \qquad Z_0 = B(0), \qquad E_B = \sum_t B(t)^2.$$

These are respectively the total four-variable constraint count, the zero-difference count, and the coupled difference energy.

4.1 Exact total variety formula

There are four ordered coupling classes. The diagonal/diagonal class contributes $(N-2)^2$. Each ordered diagonal/off-diagonal class contributes

$$(N-2)(K-1)(K-2).$$

For the off/off class, a partner of (u, v) is $(h_1/u, h_2/v)$ with $(h_1, h_2) \in H^2$. Failure to remain in D_{off} occurs only through

$$h_1 = u, \qquad h_2 = v, \qquad h_1 v = h_2 u.$$

Inclusion–exclusion gives

$$V_{\text{off}} = K^2(N-2)(N-3) - 3K(K-1)(N-3) + 2(K-1)(K-2).$$

Therefore

Theorem 4.1 (Exact $k = 4$ total)**.** *For every admissible prime N and subgroup order K,*

$$V_4 = (K^2+1)N^2 - (6K^2+3K)N + 13K^2 - 3K. \tag{3}$$

This is a finite combinatorial identity, not an asymptotic statement.

5 Exact collapse of the twist assembly

Define the correct weighted key amplitude

$$A_{\chi,\psi}(d) = \sum_{r : d_r = d} n_r \chi(u_r) \psi(v_r).$$

Double character orthogonality yields, for every t,

$$\sum_{\chi,\psi \in S} (A_{\chi,\psi} * A_{\chi,\psi})(t) = m^2 B(t). \tag{4}$$

With the unnormalized Fourier transform

$$\widehat{f}(\xi) = \sum_d f(d) e_N(-\xi d)\,,$$

this becomes

$$\sum_{\chi,\psi \in S} \widehat{A}_{\chi,\psi}(\xi)^2 = m^2 \widehat{B}(\xi). \tag{5}$$

Thus the previously mysterious "cross-twist" contribution is not an uncontrolled sum of unrelated energies; the full twist assembly is exactly one object, B.

Expanding $J^2\overline{J}^2$ gives the fourth-moment factorization

Theorem 5.1 (Exact fourth-moment reduction)**.** *For $q \in \mathbf{F}_N$,*

$$\sum_{\chi,\psi\in S} |J(\chi,\psi;q)|^4 = m^2\Sigma_4(q), \qquad \Sigma_4(q) = \sum_t B(t)e_N(qt)\,. \tag{6}$$

The same orthogonality argument extends formally to every fixed even moment, although estimating the corresponding higher-dimensional varieties is a separate problem.

6 Exact mean/variance duality

Let $V = \sum_t B(t)$ and $Z = B(0)$. Averaging the finite Fourier transform over $q \neq 0$ gives

$$\mathbf{E}_{q\neq 0}\Sigma_4(q) = \frac{NZ - V}{N-1}. \tag{7}$$

Let $\overline{B}_{\neq 0} = (V - Z)/(N-1)$. Orthogonality of additive characters yields the exact population-variance identity

Theorem 6.1 (Fourier variance duality)**.**

$$\mathrm{Var}_{q\neq 0}\,\Sigma_4(q) = N\ \mathrm{Var}_{t\neq 0}\,B(t). \tag{8}$$

Hence the desired scale

$$\mathrm{std}_{q\neq 0}\,\Sigma_4(q) = O(NK)$$

is equivalent to

$$\mathrm{std}_{t\neq 0}\,B(t) = O(K\sqrt{N}).$$

The M1 problem is therefore a concrete finite-field L^2 estimate.

7 Structural decomposition of the key distribution

Let

$$C_H(t) = \#\{(u,v) \in (H \setminus \{1\})^2 : u \neq v,\ d(u,v) = t\},$$

and let $O(t)$ be the pure off/off contribution. Then

$$B(t) = O(t) + 2(N-2)C_H(t) + (N-2)^2\mathbf{1}_{t=0}. \tag{9}$$

Because (1) never vanishes on the off-diagonal domain, $C_H(0) = 0$. Thus the variance problem separates into a generic off/off family, a lower-dimensional mixed term, and a deterministic diagonal spike.

8 Cayley transform and exact second-moment energy

8.1 Additive-energy form

For $h \neq 1$ define the Cayley transform

$$c(h) = \frac{h+1}{h-1}.$$

A direct calculation gives

$$d(u,v) = \frac{2}{c(u) - c(v)}. \tag{10}$$

Let $A = c(H \setminus \{1\})$. Since $\Delta \mapsto 2/\Delta$ is a bijection on $\mathbf{F}_N^\times$, C_H is exactly the nonzero additive-difference representation function of A, with its argument relabelled. Therefore

$$\sum_t C_H(t)^2 = E^+(A) - |A|^2. \tag{11}$$

Moreover

$$c(h^{-1}) = -c(h),$$

so $A = -A$.

8.2 Forced central-symmetry baseline

For odd K, $-1 \notin H$, hence $0 \notin A$. The involution

$$(a, b) \mapsto (-b, -a)$$

pairs representations of every nonzero difference. With $n = |A| = K - 1$, the forced nonzero energy is

$$2n(n-1) - n = (K-1)(2K-5).$$

Any excess is a genuinely new additive collision.

8.3 Cyclotomic classification for odd prime subgroup order

Assume K is an odd prime and let ζ be a primitive complex Kth root of unity. A collision

$$c(\zeta^i) - c(\zeta^j) = c(\zeta^r) - c(\zeta^s)$$

clears denominators to a sparse cyclotomic relation $P(\zeta) = 0$. Since

$$\Phi_K(X) = 1 + X + \cdots + X^{K-1}$$

is the only rational linear relation among the K powers, the collision reduces to a six-term multiset matching. The current exact audit classifies all $6! = 720$ matching systems as integer linear systems in (i, j, r, s): 30 have rank 2, 240 rank 3, and 450 rank 4. Maximal minors can lose rank only in characteristic 2. The admissible positive-dimensional families are exactly

$$(r, s) = (i, j), \qquad (r, s) = (-j, -i),$$

with all remaining rank-2/rank-3 systems forcing a zero exponent or diagonal pair.

For a non-forced normalized collision define the algebraic norm

$$R_{j,r,s} = |\mathrm{Res}(\Phi_K, P_{1,j,r,s})|, \qquad D_K = \prod R_{j,r,s}$$

(up to radical). For primes $N \equiv 1 \pmod K$, non-forced finite-field collisions occur exactly when $N \mid D_K$.

Proposition 8.1 (Exact nonexceptional $k = 2$ energy)**.** *Let K be an odd prime and $N \equiv 1 \pmod K$ a prime not dividing D_K. Then*

$$\sum_t C_H(t)^2 = (K-1)(2K-5), \qquad E^+(A) = 3(K-1)(K-2). \tag{12}$$

Consequently

$$\mathrm{Var}_{t \neq 0} C_H(t) = \frac{(K-1)(2K-5)}{N-1} - \frac{(K-1)^2(K-2)^2}{(N-1)^2}.$$

The exact exceptional sets computed by the verifier begin with

$$\mathcal{E}_{11} = \{23, 67, 89\}, \qquad \mathcal{E}_{13} = \{53, 79, 131, 157, 313\}.$$

Thus the mixed term $2(N-2)C_H(t)$ has standard deviation $O(K\sqrt{N})$ outside the explicit exceptional set.

9 The mean of the fourth-moment transform

Two deterministic N^2-scale spikes explain the observed mean. The diagonal self-coupling contributes $(N-2)^2$ to $B(0)$. In the off/off layer, $(h_1, h_2) = (1, 1)$ sends

$$(u, v) \mapsto (u^{-1}, v^{-1}), \qquad d(u^{-1}, v^{-1}) = -d(u, v),$$

so every off-diagonal key contributes, giving $(N-2)(N-3)$.

For $h_1 = h_2 = h \neq 1$, the zero-difference equation reduces to $uv = h$ and has valid count $N - 4 - \eta(h)$, where η is the quadratic character. The one-sided strata $h_1 = 1, h_2 \neq 1$ and its transpose contribute no valid zero-difference off/off points. For $h_1, h_2 \neq 1$ and $h_1 \neq h_2$, the projective completion is generically a smooth plane cubic. A direct resultant calculation gives a singularity obstruction proportional to

$$h_1^5 h_2^6 (h_1 - 1)^{14} (h_1 - h_2)^4 (h_2 - 1)^{18}.$$

Thus Hasse–Weil gives a uniform $N + O(\sqrt{N})$ count on the generic stratum after a bounded boundary correction.

Writing

$$S_\eta = \sum_{h \in H \setminus \{1\}} \eta(h),$$

one obtains the deterministic part

$$B_{0,\mathrm{det}} = 2N^2 + K^2 N - 2KN - 8N - 4K + 14 - S_\eta \tag{13}$$

and the conservative bound

$$|B(0) - B_{0,\mathrm{det}}| \leq (K-1)(K-2)(2\sqrt{N} + 28). \tag{14}$$

Combining (7), (3), and (14) yields the draft asymptotic

$$\mathbf{E}_{q \neq 0} \Sigma_4(q) = 2N^2 + O(KN + N + K^2 \sqrt{N}), \tag{15}$$

so the mean is asymptotic to $2N^2$ whenever $K = o(N^{3/4})$. The displayed resultant factorization has independent symbolic checks, but is still queued for a publication-quality hand/formal certificate.

10 The off/off elliptic pencil

10.1 Cayley coordinates

Set

$$a = c(u), \quad b = c(v), \quad \alpha = c(h_1), \quad \beta = c(h_2).$$

Then

$$T_\alpha(a) = c(h_1/u) = \frac{\alpha a - 1}{a - \alpha}, \qquad d(u, v) = \frac{2}{a - b}.$$

The fiber equation for fixed (h_1, h_2) is

$$\frac{2}{a-b} + \frac{2}{T_\alpha(a) - T_\beta(b)} = t. \tag{16}$$

After clearing denominators this is a plane cubic and, quadratically in one variable, a double cover branched at a quartic.

Let

$$p = \alpha + \beta, \qquad d = \alpha - \beta.$$

For the binary-quartic invariants I, J, the discriminant factors, up to a nonzero scalar away from characteristics 2 and 3, as

$$\Delta(t) = d^2(dt-2)^4(p-d-2)^2(p-d+2)^2(p+d-2)^2(p+d+2)^2 R_{p,d}(t), \tag{17}$$

$$R_{p,d}(t) = d^2t^4 + 2d(p^2-d^2-4)t^3 + 4(5d^2-p^2+4)t^2 - 64dt + 64. \tag{18}$$

The discriminant of $R_{p,d}$ is proportional to

$$d^2p^2(p-d-2)^2(p-d+2)^2(p+d-2)^2(p+d+2)^2.$$

For genuine Cayley parameters the last four factors never vanish. Thus the residual quartic is squarefree off the two $O(K)$ parameter strata

$$h_1 = h_2 \quad (d=0), \qquad h_1h_2 = 1 \quad (p=0).$$

10.2 Degree and parameter symmetry of the j-map

On the generic locus, $I(2/d) = 256$ and $\gcd_t(I, \Delta) = 1$. Since $\deg_t I = 4$ and $\deg_t \Delta = 8$, the j-map has degree exactly 12 and pole divisor

$$(4, 4, 1, 1, 1, 1).$$

In particular the family is non-isotrivial.

The key involution

$$(u, v) \mapsto (v^{-1}, u^{-1})$$

preserves $d(u, v)$ and sends

$$(h_1, h_2) \mapsto (h_2^{-1}, h_1^{-1}),$$

so the corresponding class arrays are identical. In Cayley coordinates this is $(\alpha, \beta) \mapsto (-\beta, -\alpha)$, which fixes d and negates p.

Conversely, the unique finite order-4 pole of $j(t)$ recovers d, and the ratio of the t^3 and t^4 Laurent coefficients at infinity recovers p^2:

$$\frac{L_3}{L_4} = \frac{4(p^2-d^2-2)}{d}.$$

Hence equality of generic same-base j-functions gives only the identity parameter or the exact involution above.

10.3 No nontrivial generic isogenies

Identical j is stronger than needed for local-system isomorphism: isogenous elliptic curves also have isomorphic rational Tate modules. The degree-12 pole structure nevertheless rules out nontrivial generic cyclic isogenies.

If a primitive separable cyclic n-isogeny existed, the two j-maps would lift through $X_0(n)$. Since

$$\psi(n) = [\mathrm{SL}_2(\mathbf{Z}) : \Gamma_0(n)] = n \prod_{q \mid n} (1 + q^{-1})$$

must divide 12, the only possibilities are

$$n \in \{2, 3, 4, 5, 6, 8, 9, 11\}.$$

At $t = \infty$, both j-maps have pole order 4. For a cusp of denominator $c \mid n$, the relevant width is

$$w(c) = \frac{n}{\gcd(c^2, n)}.$$

The two degeneracy maps are exchanged by the Fricke involution, so a lift of ramification index e would require

$$ew(c) = ew(n/c) = 4.$$

Exact enumeration of the eight candidate levels leaves equal Fricke-paired widths only at the middle cusps of $n = 4$ and $n = 9$, where $w = 1$. These would require $e = 4$, but the lift degrees are respectively 2 and 1. Thus no nontrivial separable cyclic isogeny exists. A nontrivial inseparable Frobenius factor would multiply the degree of the source j-map by a power of N, contradicting degree 12 for every member of the family.

Zarhin's geometric-field Tate theorem then implies that isomorphism of the generic rank-two ℓ-adic local systems would produce an isogeny. Hence generic local systems are geometrically non-isomorphic outside the exact involution orbit.

11 Bad fibers and trace-to-affine bookkeeping

The order-4 bad fiber occurs at

$$t_0 = \frac{2}{\alpha - \beta} = d(h_1, h_2).$$

At this value the cleared cubic factors into two lines. Thus it can contribute an N-scale point-count spike. Crucially its location is exactly the $k = 2$ Cayley difference. If $R(t)$ denotes the number of generic parameter pairs whose reducible fiber occurs at t, then

$$R(t) \le C_H(t)$$

(up to already removed special strata), and therefore

$$\frac{1}{N} \sum_t |NR(t)|^2 \le N \sum_t C_H(t)^2 = O(K^2 N) \tag{19}$$

outside the cyclotomic exceptional set.

The other four finite bad fibers are simple discriminant zeros and give irreducible nodal/rational fibers. The open key domain is obtained from the projective cubic by removing a boundary divisor of total degree 10. By Bézout, an irreducible cubic meets this boundary in at most 30 geometric points. Thus, for a generic parameter $\lambda = (h_1, h_2)$,

$$O_\lambda(t) = N - a_\lambda(t) + N\mathbf{1}_{t=d(h_1,h_2)} + e_\lambda(t), \tag{20}$$

where $a_\lambda(t)$ is the elliptic Frobenius trace represented by the middle-extension sheaf, $e_\lambda(t) = O(1)$ on good fibers, and $e_\lambda(t) = O(\sqrt{N})$ on the bounded set of remaining bad fibers. Summing the bookkeeping errors over $O(K^2)$ parameter classes contributes $O(K^4)$ to the variance.

12 Sheaf correlation input and a restricted M1 theorem draft

For a generic parameter λ, let $\pi_\lambda : E_\lambda \to U_\lambda$ be the smooth elliptic family over the good t-locus. The standard inputs line up as follows.

- Smooth proper base change and Deligne purity give a rank-two pure local system, weight zero after normalization.
- Nonconstant j and Igusa open monodromy give geometric irreducibility.
- The equations, compactification, and bad-fiber degrees are uniformly bounded; quantitative sheaf theory therefore gives absolute bounded complexity after excluding characteristics 2 and 3.
- The preceding isogeny argument and Zarhin's theorem give geometric non-isomorphism outside the exact involution orbit.
- Quantitative quasi-orthogonality then gives normalized complete cross-correlations $O(\sqrt{N})$ for non-equivalent parameter pairs, hence unnormalized elliptic-trace correlations $O(N^{3/2})$.

There are $O(K^2)$ self/involution ordered pairs and $O(K^4)$ non-equivalent pairs. After division by N to convert a complete L^2 sum into population variance, the generic elliptic traces contribute

$$\operatorname{Var}_t O(t) \ll K^2 N + K^4\sqrt{N} + K^4, \tag{21}$$

where (19) absorbs the reducible-fiber spikes. The diagonal and inverse-pair parameter strata contain only $O(K)$ classes and are no larger than the target scale under direct one-family L^2 bounds.

Restricted M1 theorem draft. Let K be an odd prime, let $N \equiv 1 \pmod{K}$ be a split prime outside the explicit cyclotomic collision set, and assume the fixed-characteristic exclusions needed by the elliptic/sheaf model. Once the algebraic model and cited sheaf-theoretic hypotheses above are certified line by line, (21), Proposition 8.1, and Cauchy–Schwarz for the covariance imply

$$K = o(N^{1/4}) \quad \Longrightarrow \quad \operatorname{Var}_{t\neq 0} B(t) = O(K^2 N).$$

Then Theorem 6.1 gives

$$\operatorname{std}_{q\neq 0} \Sigma_4(q) = O(NK), \qquad \frac{\operatorname{std} \Sigma_4}{\mathbf{E}\Sigma_4} = O(K/N),$$

using (15). This statement is not yet promoted as a final published theorem; the remaining work is verification/formalization, not a missing cancellation mechanism.

13 Finite computational certificates

The current exact $k = 4$ anchors are shown in Table 1. For $N \leq 53$, direct ordered-pair

Table 1: Independently reconstructed exact $k = 4$ invariants.

N	K	V_4	$B(0)$	$\sum_t B(t)^2$
23	11	48,621	2,609	103,156,981
53	13	423,879	12,063	3,410,502,105
199	11	4,681,821	96,641	115,589,503,389
461	23	111,147,915	641,799	26,960,554,630,421

quadruples, collapsed keys, and the fast H^2 partner algorithm agree entry-by-entry on $B(t)$. Python and an independent C++ implementation also agree on direct Jacobi fourth moments. The twist-collapse identity is checked through $N = 199$ in both time and Fourier domains; the C++ version uses direct DFT/circular convolution rather than NumPy FFT.

Corrected finite designs also show the expected scaling trend: These data suggest $\mathbf{E}\Sigma_4 \sim$

Table 2: Corrected numerical scaling of Σ_4; evidence only.

N	K	m	$\mathbf{E}(\Sigma_4)/N^2$	$\mathrm{std}(\Sigma_4)/(NK)$
199	11	18	1.8556	3.7996
461	23	20	1.8895	3.8797
521	13	40	1.9357	3.8408

$2N^2$ and $\mathrm{std}\,\Sigma_4 \sim 4NK$, but the constants are not promoted from numerical evidence.

14 Historical falsifiers and negative results

A reliable research workspace must preserve failed routes without allowing them to masquerade as current proofs.

14.1 The historical Möbius-fiber assembly

Eliminating (x, y) from the off-diagonal inverse formulas produces

$$v_d(u) = \frac{(d-1)u + 1}{d + 1 - u}.$$

Without saturation this eliminated equation contains $(u, v) = (1, 1)$ for every d, although the inverse pair map is undefined there. The historical graph therefore contains a phantom point. Exact comparison with the true key table gives:

- for $d \neq 0$, $A_{\mathrm{hist}} = A_{\mathrm{key}} + 1$ for every twist pair;
- for $d = 0$ with first twist trivial, both equal $N - 2$;
- for $d = 0$ with first twist nontrivial, $A_{\mathrm{hist}} = -1$ while $A_{\mathrm{key}} = N - 2$.

Thus the old proof cannot be repaired merely by adding cross terms or strengthening a Weil estimate.

14.2 Ordered-coupling bookkeeping bug

An archived large-m producer included diagonal→off coupling but omitted the reverse ordered off→diagonal class. At $N = 53, K = 13$ the missing class has exact size

$$(N-2)(K-1)(K-2) = 51 \cdot 132 = 6732,$$

which explains the historical difference $423879 - 417147 = 6732$. Current verifiers compare the entire $B(t)$ distribution and permanently guard against this recurrence.

14.3 Other quarantined claims

The workspace also quarantines a historical spacing test that compared consecutive-gap ratio statistics with an ordinary GOE spacing CDF, a word-trace program that terminated with an unresolved word, and a Hecke-centring numerical line whose selected subspace and random-centering construction did not represent the claimed Hecke eigenspace mechanism. These failures are research information, not evidence for BGS.

15 Reproducible computation and language architecture

The current implementation deliberately avoids a one-script-per-run layout.

- Python is used for symbolic/exact research glue, finite-field analysis, deterministic probes, and result assembly.
- C++20 is used for expensive finite-field loops, exact dynamic programming, and independent exhaustive verification.
- A second language implementation is kept only when independence is itself part of a scientific certificate.
- Historical Julia sources remain once in the immutable provenance archive. They are mapped to consolidated supported components rather than mechanically translated one for one.

The migration audit found, among other things, that an old Perron-transfer script had an impossible spectral-radius bracket because a zero-cost symbol induces a weight-one self-loop, and that a helper named "projective order" actually tested return to the identity matrix. These are examples of why language migration is treated as a scientific audit rather than syntax conversion.

16 What remains before a local BGS theorem

Even a fully certified restricted M1 theorem is not a local spacing theorem. The next stages are:

1. finish publication-level certification of the $K = o(N^{1/4})$ fourth-moment concentration statement;
2. identify the precise spectral observable/local window for which the moment information is sufficient;
3. control higher moments or a Stieltjes/resolvent object strongly enough to obtain a local law;
4. remove residual arithmetic symmetries and state the unfolding procedure exactly;
5. prove the relevant random-matrix comparison or universality step;
6. use numerical GOE controls only as falsifiers and calibration, never as substitutes for the proof.

The general BGS conjecture remains outside the current theorem set.

17 Conclusion

The central methodological change is the replacement of a fragile twist-by-twist Möbius argument by an exact pair-key distribution. This turns the fourth moment into a single finite Fourier transform with an exact variance duality. The mixed contribution is solved in a substantial arithmetic regime by Cayley energy and cyclotomic collision classification, while the remaining off/off contribution is an explicit bounded-complexity elliptic family. The first plausible asymptotic concentration range is therefore no longer speculative: it reduces to a short list of certifiable algebraic and sheaf-theoretic steps. The project nevertheless keeps a strict distinction between exact finite identities, theorem drafts, numerical evidence, and the still-open local BGS conclusion.

A Verification checklist for the restricted M1 draft

Before the draft proposition is promoted, the repository requires all of the following:

1. a hand or formal certificate for the generic cubic singularity resultant;
2. a hand or formal certificate for the binary-quartic discriminant, $\gcd(I, \Delta) = 1$, and the degree-12 j-map data;
3. a written cusp-width proof of the no-isogeny lemma, with characteristic hypotheses stated explicitly;
4. exact theorem references and hypotheses for Igusa monodromy, Zarhin's geometric-field Tate theorem, Deligne purity, and quantitative quasi-orthogonality;
5. a projective-model proof of the trace-to-affine formula (20), including every boundary/bad fiber;
6. automated regression gates for all finite identities used as normalization checks.

B Selected references

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